

AI Mental Health Companion

ABSTRACT

Mental health matters more than ever these days. People face all kinds of emotional struggles—stress, anxiety, depression, and that constant pressure from school, work, social expectations, or their own personal problems. But getting real help isn’t always easy. Stigma still hangs around, not enough folks know what support looks like, therapy costs too much, and honestly, there just aren’t enough trained professionals to go around. So, a lot of people hesitate or just can’t get help when they actually need it. That’s where technology steps up. The AI Mental Health Companion aims to fill in those gaps, offering emotional support and guidance through artificial intelligence. It’s basically a smart assistant you can talk to—type out your thoughts or just speak them, and it listens. Using things like Natural Language Processing, machine learning, and sentiment analysis, it’s able to pick up on your emotional state—whether you’re feeling stressed, down, happy, or anxious. Then, it responds with encouragement, coping advice, motivation, or even suggestions for how to relax. One real bonus? It’s always available. No need for appointments or to worry about office hours—this digital assistant is there 24/7. If you’re feeling overwhelmed in the middle of the night, it’ll still respond. Plus, the platform’s private and non-judgmental, so you can be honest about what you’re feeling without worrying about being criticized or embarrassed. That helps a lot for people who struggle with the stigma around mental health. The system does more than just listen, though. There’s a mood tracker that lets you keep tabs on how you’re feeling over time. It looks for patterns, and then tailors its suggestions to help you find more balance. You also get access to guided meditations, breathing exercises, motivational messages, and wellness tips—all designed to help you manage stress and stay positive. The AI Mental Health Companion isn’t meant to replace real, professional therapy. It’s a supplement, offering support until—or alongside—professional help. And if it spots signs of serious distress, it will encourage you to reach out to a qualified mental health worker. All in all, this Companion is an innovative way to boost mental health awareness and offer accessible support. It blends artificial intelligence, easy-to-use conversations, and wellness resources to create a safe space for sharing emotions, getting advice, and learning to cope better. It helps bridge the gap that so many people face when traditional mental health services are out of reach.

CONTENT

1. Introduction

1.1 Introduction.....	1
1.2 Scope of the Study.....	2
1.3 Objective.....	4
1.4 Overview of the project.....	5
1.5 Problem Statement.....	7

2. System Analysis

2.1 Existing System.....	9
2.1.1 Disadvantages of Existing System.....	11
2.2 Proposed System.....	12
2.2.1 Advantages of proposed system.....	13

3. System Specification

3.1 Introduction.....	14
3.1.1 Hardware Requirements.....	15
3.1.2 Software Requirements.....	15
3.2 Software Description.....	16
3.2.1 Visual Studio Code.....	16
3.2.2 PyCharm.....	17
3.2.3 React.....	18
3.2.4 Node.js.....	19
3.2.5 Express.js.....	21
3.2.6 MongoDB.....	22
3.2.7 Mongoose.....	23
3.2.8 JSON Web Token (JWT).....	25
3.2.9 bcrypt.....	26
3.2.10 TensorFlow.js.....	27

3.2.11 Role-Based AI System.....	28
3.2.12 Stress Prediction Model.....	29
3.2.13 Multer.....	30
3.2.14 Natural Language Processing Libraries (NLTK/spaCy)	31
3.2.15 TensorFlow/Deep Learning Models.....	32
3.2.16 MySQL Database.....	36
3.2.17 Web Browser.....	40
4. System Design	
4.1 Introduction.....	43
4.2 System Architecture Design.....	44
4.2.1 Explanation.....	45
4.3 Input Design.....	46
4.3.1 Features of Input Design.....	47
4.4 Output Design.....	48
4.4.1 Features of Output Design.....	49
4.5 Data Flow Diagram.....	50
4.5.1 Explanation.....	52
4.5.2 Level of DFD.....	54
5. System Implementation	
5.1 Introduction.....	59
5.1.1 Implementation Modules.....	60
5.2 Result and Reports.....	75
5.3 Screenshots.....	146
6. Conclusion and Future Enhancement	
6.1 Conclusion.....	156
6.2 Future Enhancement.....	158
References.....	162